# MD. SADIK HASAN RAFI

## SOFTWARE ENGINEER

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Web developer with 2.5+ years' experience building business web apps in React.js, Next.js, Node.js and PostgreSQL/MySQL. I learn how a client's business runs, turn it into working software and explain it in plain language. Hands-on with REST APIs, auth, debugging and AI-assisted development.

### PROFESSIONAL EXPERIENCE

**Celloscope (Intern)** **April 2023 - October 2023**

- Developed frontend features using Angular and TypeScript for enterprise applications.
- Contributed to production systems including ABS Portal and Ampere Portal.
- Collaborated with senior engineers in an Agile development environment.

**Celloscope (Junior Software Engineer)** **October 2023 - July 2025**

- Built authentication and authorization with JWT and role-based access control (RBAC) across enterprise portals, including ABS Portal, Ampere Portal and SMS Alert.
- Built a Maker-Checker approval workflow (Create → Submit → Approve) for merchant account operations.
- Connected REST APIs to server-side logic in an Angular, Spring Boot, RabbitMQ and PostgreSQL stack.
- Added real-time updates and notifications using WebSocket (STOMP).
- Added audit trails and centralized error logging to support monitoring and debugging.
- Built reporting features with PDF generation and CSV/Excel export using jsPDF and html2canvas.
- Added English and Bengali language support using Angular i18n.
- Worked in Agile sprints, using Git/GitHub for version control and team collaboration.

### PROJECTS

1. **E-China | Logistics & Import Management Platform** **Live link**

Developed a full-stack system to manage product sourcing from China, import logistics to Bangladesh, and local sales operations.

- Built JWT authentication with RBAC, enforcing permission boundaries across supplier, admin and sales roles.
- Designed REST APIs and PostgreSQL models with Prisma for procurement, inventory and sales relationships.
- Integrated frontend and backend through typed API contracts, improving data consistency across modules.
- Built an operational dashboard in Next.js and shadcn/ui for import pipelines, procurement status and sales.
- Managed the procurement lifecycle data flow from order creation to import tracking and local delivery.

Tech Stack: Next.js, TypeScript, Tailwind CSS, shadcn/ui, Redux, Node.js, Express, Prisma, PostgreSQL.

2. **Village Agro Farm | Agricultural ERP & Accounting System**

An ERP that merges farm operations, money management and accounting, with subscription billing.

- Built accounting modules: invoices, bills, general ledger, trial balance, balance sheet, tax summary and bank reconciliation.
- Implemented subscription billing with bKash Tokenized Checkout, including grant-token lifecycle, agreement and payment flows, and webhook handlers.
- Normalized the database schema, wrote seed data and completed a full ORM migration from Prisma to Drizzle.
- Built PDF, CSV and Excel export features.
- Developed React/TypeScript screens for transaction history, projects, checkout/billing and settings.

Tech Stack: React, TypeScript, Tailwind CSS, Express, Drizzle ORM (migrated from Prisma), MySQL.

**3. Pink Life Line | Menstrual Health Tracking Platform** **Live link**

- Architected a modular Express and MongoDB backend for a women's health platform, combining period tracking, doctor tele-consultation booking, community forums and self-assessment tools in one system.
- Designed a role-based access control (RBAC) system with dynamic, database-driven route permissions per admin role, allowing access rules to change without redeployment.
- Built a doctor scheduling system with recurring weekly availability, holiday overrides and slot-capacity limits, driving real-time appointment booking, confirmation and cancellation for patients and admins.
- Implemented Firebase Cloud Messaging notifications with a cron-based scheduler to auto-send personalized cycle reminders based on predicted dates.
- Developed a Reddit-style community module with nested comments, independent upvote/downvote logic, saved posts, search and admin content moderation.
- Integrated Cloudinary, Google Cloud Vision and JWT/bcrypt auth for media handling, doctor credential verification and secure multi-role login flows.

**4. Liguni | Retail POS & Shop Management System**

- Building a POS for a footwear retailer with variant-based inventory, barcode generation and receipt printing.
- Integrated SSLCommerz payments plus bKash, card, cash and COD orders through the Steadfast courier API.
- Implemented admin/salesman role-based access, WhatsApp order notifications and a mini-CRM.
- Work directly with the shop owner on requirements; wrote the proposal and contract terms.

Tech Stack: React, TanStack Query, Tailwind CSS, shadcn/ui, Express, TypeScript, Prisma, PostgreSQL.

## AREA OF EXPERTISE

Languages: JavaScript, TypeScript, HTML5, CSS3
Frontend: React.js, Next.js, Angular, Tailwind CSS
Databases: PostgreSQL, MySQL, MongoDB
DevOps: Linux, SSH, Docker, Git/GitHub
AI Tools: Claude, Cursor, Copilot, GPT, Gemini

Backend: Node.js, Express.js, NestJS, REST APIs
Security: JWT, bcrypt, RBAC, API authentication
ORM / ODM: Mongoose, Prisma, Drizzle
Real-time: Socket.IO, WebSocket (STOMP), RabbitMQ
Integrations: bKash, SSLCommerz, Steadfast, FCM

## ACHIEVEMENTS

- 1st Place, University Business Idea Contest
- 3rd Place, University Hackathon Competition
- Ranked 49th among 200+ participants in Programming Contest

## EDUCATION & COURSES

**B.Sc. in Computer Science and Engineering**

Daffodil International University
2019 - 2023 | CGPA 3.44

**Courses**

- Complete Web Development - Jhankar Mahbub
- Learning & Earning Development Project (LEDP)